

^{35}Ar $\varepsilon+\beta^+$ decay (1.7755 s) 1980Wi13,1984Ad01

Parent: ^{35}Ar : $E=0$; $J^\pi=3/2^+$; $T_{1/2}=1.7755$ s 14; $Q(\varepsilon)=5966.2$ 7; $\%\varepsilon+\%\beta^+$ decay=100.0

^{35}Ar - J^π , $T_{1/2}$: From Adopted Levels of ^{35}Ar .

^{35}Ar - $Q(\varepsilon)$: From 2021Wa16.

1980Wi13: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 11.0-MeV protons from the ONR-CIT tandem accelerator impinging on LiCl and PbCl₂ wrapped in 25- μm Ta foil. γ rays were detected using a 100-cm³ Ge(Li) detector. Measured E_γ and I_γ for 20 γ -ray peaks. Deduced decay branching ratios for 9 levels, and log ft . Compared with shell-model calculations.

1984Ad01: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 13-MeV protons from the University of Washington FN tandem accelerator impinging on a thick KCl target. γ rays were detected using a 28% Ge(Li) detector. Measured E_γ and I_γ for 8 γ -ray peaks. Deduced decay branching ratios for 6 levels, and log ft . Compared with shell-model calculations.

1985Da04: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 9-MeV protons from the Van de Graaff accelerator at the University of Pittsburgh impinging on a NaCl target. γ rays were detected using a a Ge(Li) detector with NaI anti-Compton shielding. Measured E_γ and I_γ for 4 γ -ray peaks. Deduced 3 levels.

1979Ha04: ^{35}Ar was produced via the V(p,6p11n) spallation reaction with 600-MeV protons from the CERN synchrocyclotron bombarding a 38⁻ g/cm² vanadium target. γ rays were detected using a Ge(Li) detector. Measured E_γ and I_γ for the 1219 γ . Deduced the decay branching ratio for the 1219-keV level.

1977Az01: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-13-MeV protons from the McGill synchrocyclotron impinging on CCl₄ and LiCl targets and 8-MeV protons from the University of Montreal impinging a thick LiCl pellet. Positrons were detected using a plastic (NE-102) coupled to an RCA 6342A photomultiplier tube. γ rays were detected using a 18% Ge(Li) detector. Measured E_γ and I_γ for 3 γ -ray peaks. Deduced decay branching ratios for 3 levels and parent ^{35}Ar $T_{1/2}$. 1978Az01: $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction threshold.

1971Ge04: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-MeV protons from the Chalk River tandem accelerator impinging on PbCl₂ targets. γ rays were detected using two coaxial Ge(Li) detectors. Measured E_γ and I_γ for 4 γ -ray peaks. Deduced 4 levels, decay branching ratios, log ft , and parent ^{35}Ar $T_{1/2}$.

1971De05: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction on KCl targets. γ rays were detected using a 25 cm³ Ge(Li) detector. Measured E_γ and I_γ . Deduced 6 levels, decay branching ratios, and log ft .

1969Wi18,1969Fr09: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with a proton beam from the Harwell tandem accelerator impinging on 3.75-mg/cm² KCl deposited on 0.13-mm Au backings and 160-mmHg gas CCl₄ targets. γ rays were detected using a Ge(Li) detector. Measured E_γ and I_γ for 2 γ -ray peaks. Deduced 2 levels, decay branching ratios, (p,n) reaction threshold and β end-point energy, log ft , and parent ^{35}Ar $T_{1/2}$.

1960Wa04: ^{35}Ar was produced at Crocker Laboratory and the Lawrence Radiation Laboratory. β spectra were measured with a uniform-held 180 $^\circ$ -deflection single-focusing spectrometer. Deduced Coulomb-energy differences between mirror ground-state transitions.

1956Ki29: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-MeV protons from the BNL 60-inch cyclotron impinging on a liquid CCl₄ target. β particles were detected using a conventional thin-lens, iron-free, magnetic spectrometer. γ rays were detected using a NaI(Tl) crystal scintillation counter. Measured E_β and total I_β for 2 positron decays to ^{35}Cl excited states, E_γ and I_γ for 2 γ -ray peaks. Deduced 2 levels and decay branching ratios.

^{35}Ar β asymmetry parameter measurements: 1993Co07, 1991Na05, 1991Mi18, 1988Ga02, 1965Ca04.

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$
0	3/2 ⁺	stable
1219.2 2	1/2 ⁺	0.119 ps 14
1763.1 2	5/2 ⁺	0.36 ps 3
2645.1 10	7/2 ⁺	0.159 ps 21
2693.8 2	3/2 ⁺	32 fs 6
3002.3 3	5/2 ⁺	22 fs 6
3918.4 5	3/2 ⁺	4.9 fs 14
3967.7 4	1/2 ⁺	11.8 fs 28
4624.5? 5	3/2,5/2	40 fs 17

[†] From a least-squares fit to γ -ray energies.

[‡] From the Adopted Levels.

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ε, β^+ radiations						Comments
E(decay)	E(level)	$I\beta^+ \ddagger$	$I\varepsilon \ddagger$	Log ft	$I(\varepsilon+\beta^+) \ddagger\ddagger$	
(1341.7 [#] 13)	4624.5?	$<8.0 \times 10^{-4}$	<0.0035	>3.8	<0.0043	$I(\varepsilon+\beta^+)$: weighted average of 1.223 46 (1969Wi18,1969Fr09), 1.22 20 (1971Ge04), 1.55 15 (1977Az01), 1.34 8 (1979Ha04), 1.23 7 (1984Ad01), 1.228 30 (1980Wi13). $I(\varepsilon+\beta^+)$: deduced from $100 - \Sigma I(\varepsilon+\beta^+)$ (excited states), where $\Sigma I(\varepsilon+\beta^+)$ (excited states)=1.79 6. Level scheme fed by β^+ decay may be complete considering the highest observed level is at 4.6 MeV. Observed $\Sigma I(\varepsilon) \approx 0.08$. Evaluators assume $\Sigma I(\varepsilon) \leq 0.08$ for all unobserved ε feedings to high-lying levels and include this limit in the uncertainty for $\Sigma I(\varepsilon+\beta^+)$ (excited states) and $I(\varepsilon+\beta^+)$ (gs).
(1998.5 13)	3967.7	0.015 11	0.0016 10	4.5	0.017 11	
(2047.8 13)	3918.4	0.0097 34	8.4×10^{-4} 27	4.8	0.0105 34	
(2963.9 13)	3002.3	0.090 15	0.0011 2	5.0	0.091 15	
(3272.4 13)	2693.8	0.166 10	0.00127 8	5.0	0.167 10	
(3321.1 16)	2645.1	<0.00149	$<1.1 \times 10^{-5}$	>7.1	<0.0015	
(4203.1 13)	1763.1	0.259 16	7.1×10^{-4} 5	5.5	0.260 16	
(4747.0 13)	1219.2	1.24 6	0.0021 1	5.10 2	1.24 6	
(5966.2 16)	0	98.14 10	0.0731 8	3.760 1	98.21 10	

[†] Deduced from γ -ray intensity balance, relative to $I_\gamma(1219\gamma)=100$, normalized to absolute $\%I_\gamma(1219\gamma)=1.26$ 5.

[‡] Absolute intensity per 100 decays.

[#] Existence of this branch is questionable.

 $\gamma(^{35}\text{Cl})$

I_γ normalization: absolute $\%I_\gamma(1219\gamma)=1.26$ 5 deduced from weighted average of 1.223 46 (1969Wi18,1969Fr09), 1.22 20 (1971Ge04), 1.55 15 (1977Az01), 1.34 8 (1979Ha04), 1.23 7 (1984Ad01), 1.228 30 (1980Wi13).

$E_\gamma \dagger$	$I_\gamma \ddagger \&$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
916 ^a		3918.4	3/2 ⁺	3002.3	5/2 ⁺			The intensity could not be determined due to being obscured by another peak (1980Wi13).
930.7 2	1.57 20	2693.8	3/2 ⁺	1763.1	5/2 ⁺	M1+E2	+0.09 3	$\%I_\gamma=0.0198 +34-32$ I_γ : weighted average of 1.13 26 (1980Wi13) and 1.66 12 (1984Ad01).
965 ^a	$<0.87 @$	3967.7	1/2 ⁺	3002.3	5/2 ⁺			$\%I_\gamma<0.0110$
1219.3 2	100.0 15	1219.2	1/2 ⁺	0	3/2 ⁺	M1+E2	0.093 10	$\%I_\gamma=1.26$ 6 I_γ : others: 100.0 30 from 1984Ad01, 100 10 from 1977Az01, 100 from 1969Wi18, 1971Ge04, 1979Ha04, and 1985Da04.
1225 ^a		3918.4	3/2 ⁺	2693.8	3/2 ⁺			The intensity could not be determined due to being obscured by another peak (1980Wi13).
1239 ^a	$<0.30 @$	3002.3	5/2 ⁺	1763.1	5/2 ⁺	[M1,E2]		$\%I_\gamma<0.0038$
1274 ^a	$<0.42 @$	3967.7	1/2 ⁺	2693.8	3/2 ⁺	[M1,E2]		$\%I_\gamma<0.0053$
1474.8 3	1.02 11	2693.8	3/2 ⁺	1219.2	1/2 ⁺	M1+E2	-0.63 21	$\%I_\gamma=0.0129$ 19 I_γ : weighted average of 0.83 13 (1980Wi13) and 1.08 7 (1984Ad01).

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^{35}Ar $\varepsilon + \beta^+$ decay (1.7755 s) **1980Wi13,1984Ad01** (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_\gamma^\dagger$	$I_\gamma^{\ddagger\&}$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
1763.0 2	22.7 4	1763.1	5/2 ⁺	0	3/2 ⁺	M1+E2	-2.85 10	$\%I_\gamma = 0.286$ 13 I_γ : weighted average of 23.4 13 (1969Wi18), 20.0 27 (1971Ge04), 22 6 (1977Az01), 23.08 37 (1980Wi13), 21.85 67 (1984Ad01), and 21.7 9 (1985Da04).
1783 ^a	<0.39 [@]	3002.3	5/2 ⁺	1219.2	1/2 ⁺	[E2]		$\%I_\gamma < 0.0049$
1931 ^a	<0.21 [@]	4624.5?	3/2,5/2	2693.8	3/2 ⁺			$\%I_\gamma < 0.0027$
2155.1 15	0.15 6	3918.4	3/2 ⁺	1763.1	5/2 ⁺	[M1,E2]		$\%I_\gamma = 0.0019$ +9-8 I_γ : from 1984Ad01. Other: <0.22 (1980Wi13).
2204 ^a	<0.27 [@]	3967.7	1/2 ⁺	1763.1	5/2 ⁺	[E2]		$\%I_\gamma < 0.0034$
2645 ^a	<0.12 [@]	2645.1	7/2 ⁺	0	3/2 ⁺	E2		$\%I_\gamma < 0.00151$
2693.7 2	10.96 18	2693.8	3/2 ⁺	0	3/2 ⁺	M1+E2	+0.26 2	$\%I_\gamma = 0.138$ 6 I_γ : weighted average of 10.2 15 (1971Ge04), 10.96 18 (1980Wi13) and 11.02 34 (1984Ad01).
2699 ^a	<0.33 [@]	3918.4	3/2 ⁺	1219.2	1/2 ⁺	[M1,E2]		$\%I_\gamma < 0.0042$
2748.5 6	0.477 34	3967.7	1/2 ⁺	1219.2	1/2 ⁺	[M1]		$\%I_\gamma = 0.0060$ 5 I_γ : weighted average of 0.48 6 (1980Wi13) and 0.476 34 (1984Ad01).
2861 ^a	<0.099 [@]	4624.5?	3/2,5/2	1763.1	5/2 ⁺			$\%I_\gamma < 0.00125$
3002.1 4	7.29 14	3002.3	5/2 ⁺	0	3/2 ⁺	M1+E2	+0.08 2	$\%I_\gamma = 0.092$ 4 I_γ : weighted average of 9.0 15 (1971Ge04), 7.24 14 (1980Wi13), 7.45 23 (1984Ad01) and 6.9 5 (1985Da04).
3405 ^a		4624.5?	3/2,5/2	1219.2	1/2 ⁺			I_γ : could not be determined due to being obscured by another peak (1980Wi13).
3918	0.52 3	3918.4	3/2 ⁺	0	3/2 ⁺	M1+E2		$\%I_\gamma = 0.0066$ 5
3968	0.106 25	3967.7	1/2 ⁺	0	3/2 ⁺	[M1,E2]		$\%I_\gamma = 0.0013$ 4
4624 ^a	<0.035 [@]	4624.5?	3/2,5/2	0	3/2 ⁺	D+Q		$\%I_\gamma < 4.4 \times 10^{-4}$

[†] From 1984Ad01, unless otherwise noted.[‡] from 1980Wi13, unless otherwise noted.[#] From Adopted Gammas.[@] 1980Wi13 gives upper limits of I_γ at 98% confidence level. These peaks are not observed in the spectra of 1980Wi13 or in 1984Ad01.[&] For absolute intensity per 100 decays, multiply by 0.0126 5.^a Placement of transition in the level scheme is uncertain.

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Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
 — $I_\gamma < 10\% \times I_\gamma^{\max}$
 — $I_\gamma > 10\% \times I_\gamma^{\max}$
 - - - γ Decay (Uncertain)

Decay Scheme

Intensities: I_γ per 100 parent decays